

ATS Resume Scoring System – Final Project Report

Project Overview

The ATS Resume Scoring System is an AI-powered application that evaluates resumes against job descriptions using NLP techniques. It generates ATS scores, improvement suggestions, and professional reports.

System Workflow

- Select job role and input job description
- Upload one or more resume PDFs
- Extract resume text
- Apply NLP preprocessing
- Calculate semantic similarity scores
- Generate skill gap analysis
- Rank resumes
- Generate professional PDF report

Technologies Used

Technology	Purpose
Python	Core logic
Streamlit	User Interface
spaCy & NLTK	NLP Processing
Sentence Transformers	Semantic Similarity
Matplotlib	Charts & Diagrams
ReportLab	PDF Generation

Objectives

- Automate resume screening using NLP technique
- Compare resumes with job descriptions semantically
- Generate an ATS compatibility score (0–100)
- Perform section-wise resume evaluation
- Identify missing skills and skill gaps
- Provide actionable resume improvement suggestions
- Rank multiple resumes based on ATS score
- Generate a professional, downloadable PDF report

Step-by-Step Execution Process

- Install required Python libraries
- Run Streamlit application
- Select job role and paste job description
- Upload resume files
- Run ATS analysis
- View and download reports

Final Output

- Overall ATS score (0–100)
- Section-wise score breakdown
- Skill gap analysis
- Resume strengths
- Improvement suggestions
- Resume ranking
- Teal-colored horizontal bar chart
- Professional PDF report

Advantages

- ATS-friendly resume evaluation
- Multiple resume ranking
- Skill gap identification
- Professional PDF reporting
- Cost-effective ATS simulation tool
- User-friendly UI

Limitations

- Dependent on resume text quality
- Limited language support
- Simulated ATS logic

Future Enhancements

- Multilingual resume support
- Advanced ML skill extraction
- Real ATS integration
- Cloud deployment

Conclusion

The **ATS Resume Scoring System** successfully demonstrates how AI and NLP can be applied to automate resume evaluation. By combining semantic similarity, skill gap analysis, ranking, and professional reporting, this project closely resembles real-world ATS platforms and provides valuable insights for job seekers.

ATS Resume Scoring System - Process Flow Diagram

